

Supplementary Materials for
**Seismic and mineralogical evidence for an iron-rich mega–ultralow-velocity
zone beneath Hawai’i**

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The PDF file includes:

Figs. S1 to S12
Tables S1 and S2
Legends for data S1 and S2
References

Other Supplementary Material for this manuscript includes the following:

Data S1 and S2

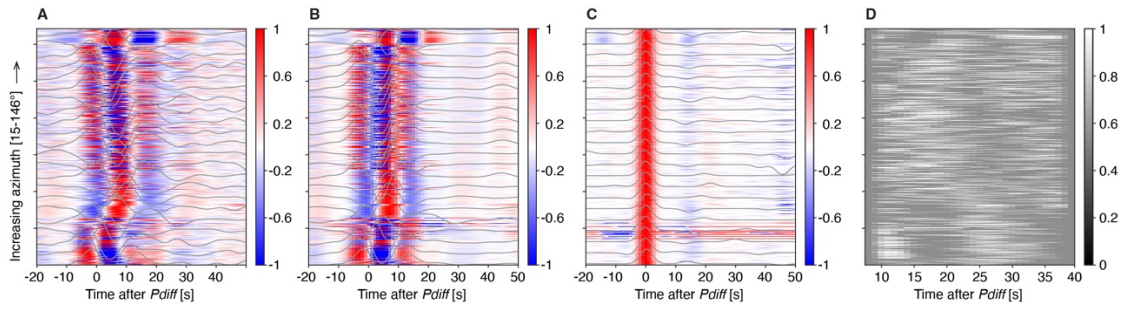


fig. S1. Processing of Pdiff waveforms. (A) Raw vertical displacement seismograms and (B) synthetic waveforms computed for PREM. The red-blue color scale shows normalized amplitude relative to the main Pdiff arrival. (C) Source deconvolved seismograms obtained by taking (A) and deconvolving with (B). (D) Waveforms in (C) applied with the histogram equalization, used as input for the Sequencer.

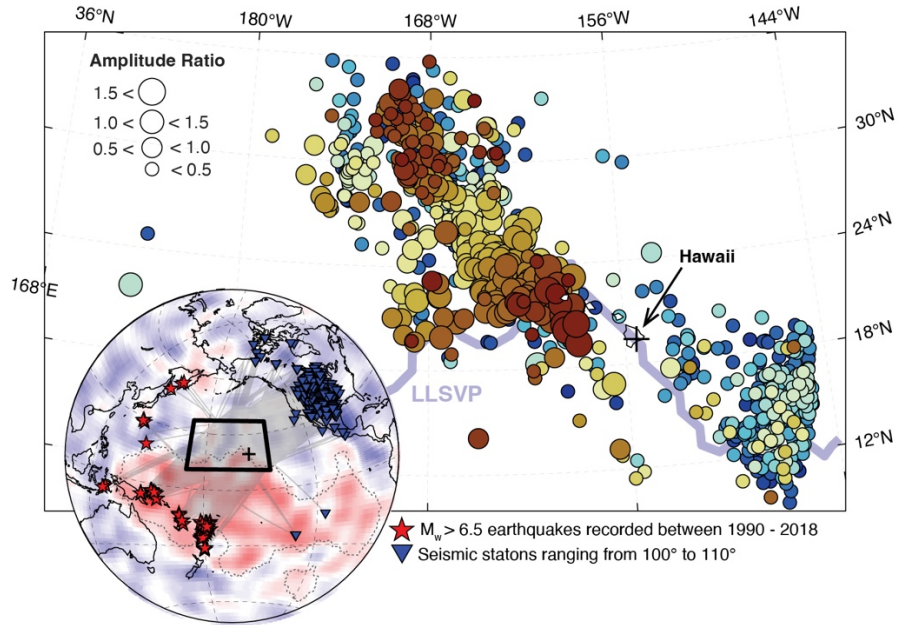


fig. S2. Spatial distribution of the Sdiff postcursors in Hawai'i (after (12)). Inset map shows the locations of earthquakes (red) and seismic stations (blue) used in the study, along with the geographic extent of the Pacific LLVP, marked by the transition from red to blue regions.

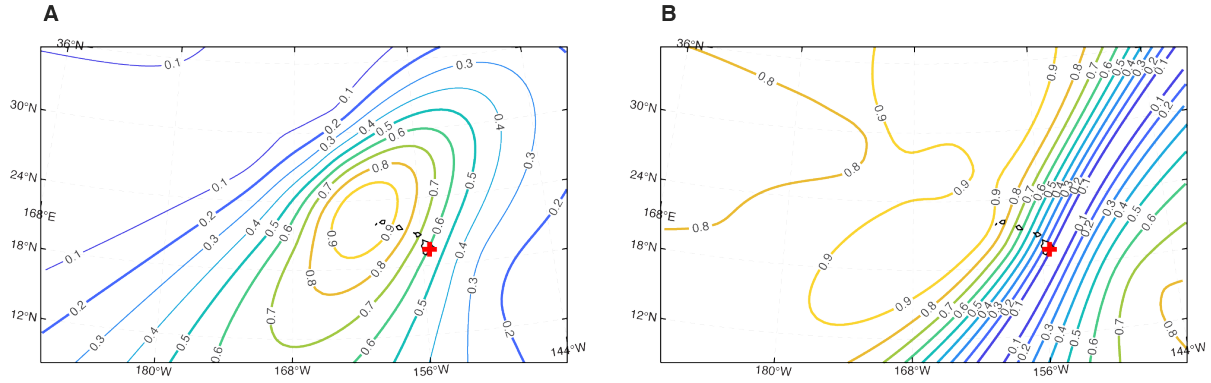


fig. S3. Correlation coefficients between sequenced index and midpoint distance from putative mega-ULVZ locations. Contour plot of the correlation between sequenced index and path-midpoint distance associated with each potential location of the mega-ULVZ for combined Pdiff (A) and Sdiff (B) datasets. At each location, distance is computed to all the midpoints for each diffracted path in the dataset and then the correlation coefficient is computed between that distance and the sequenced index. The product of the Pdiff and Sdiff correlation coefficients is shown in Fig. 2A). For these computations, only data exhibiting clear postcursors (the latter half of the indices) were included, while waveforms without postcursor signals were excluded.

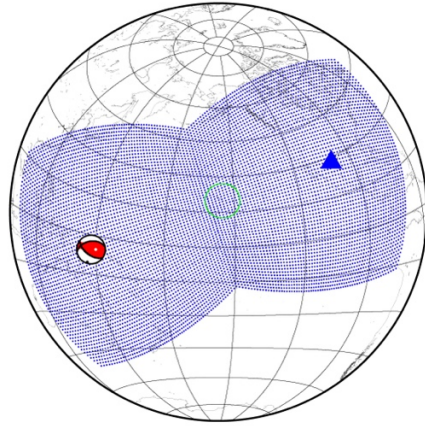


fig. S4. SPECTFEM-3D modeling domain. SPECTFEM-3D Globe spectral element grids, illustrating the model domain with two chunks covering the major source and receiver regions relevant to the dataset. The reference source parameters are based on the CMT solution (53) of M_w 6.9 earthquake in the Eastern New Guinea region on April 17, 2012, with a hypocentral depth of 208.6 km. Multiple point-profile models of the ULVZ are tested using the 1-D transversely isotropic PREM as the background. The primary synthetic waveforms and analyses are derived from SPECTFEM-3D outputs, verified with additional simulations performed using AXISSEM-3D (e.g., Fig.S6-S8.)

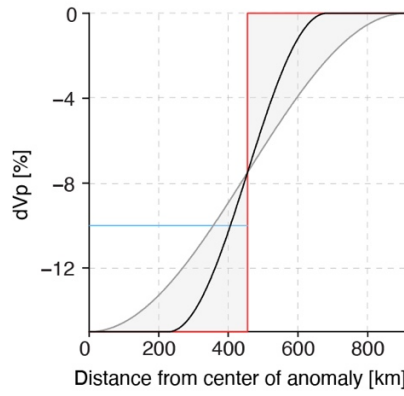


fig. S5. ULVZ models with different edge sharpness in velocity. The effect of velocity gradient sharpness at the edges of ULVZ models is tested by applying a cosine taper to create smooth edges. The shaded region illustrates the transition from abrupt to smooth model boundaries, with red and blue lines representing synthetic measurements corresponding to those plotted in Fig. 3A.

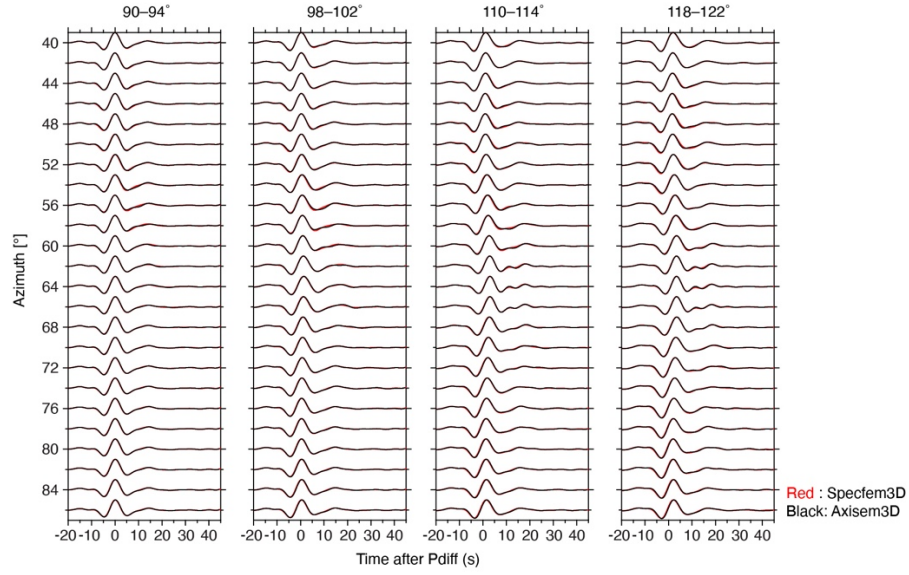


fig S6. Synthetic Pdiff waveforms. Comparison of synthetic waveforms computed based on two different spectral element methods employed in this study. The model includes a cylindrical ULVZ structure with $H = 50$ km, $W = 910$ km, $dVp = -10\%$, and $dVs = -20\%$, embedded within the PREM background. The epicentral distance ranges of the receivers that record the synthetic seismograms are shown above each panel.

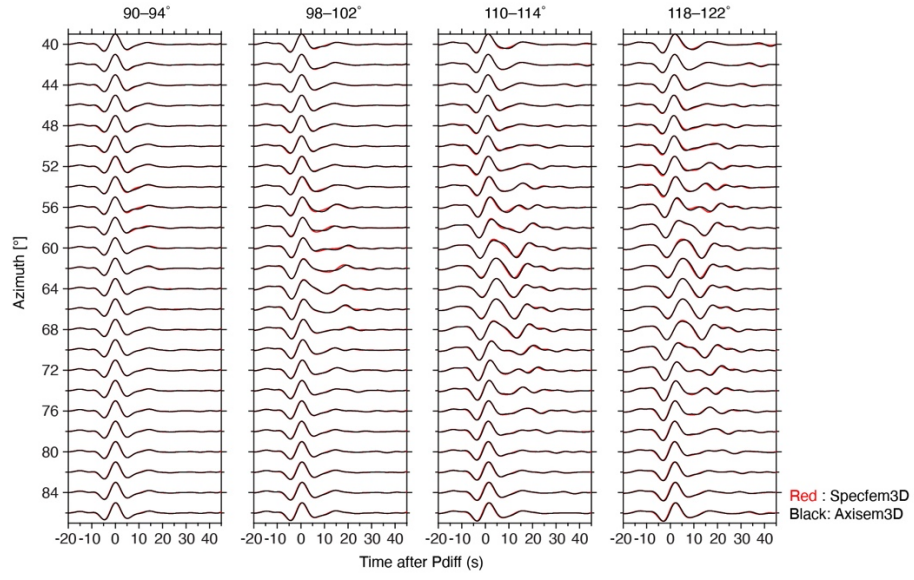


fig. S7. Synthetic Pdiff waveforms. Same as fig. S6 but using a $dVp = -20\%$.

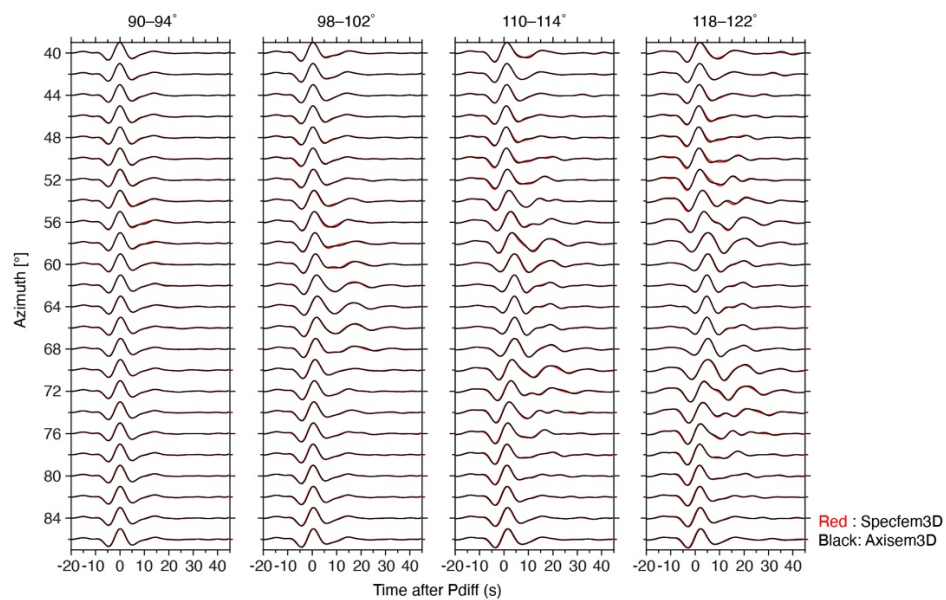


fig. S8. Synthetic Pdiff waveforms. Same as fig. S6 but using a $H = 100$ km.

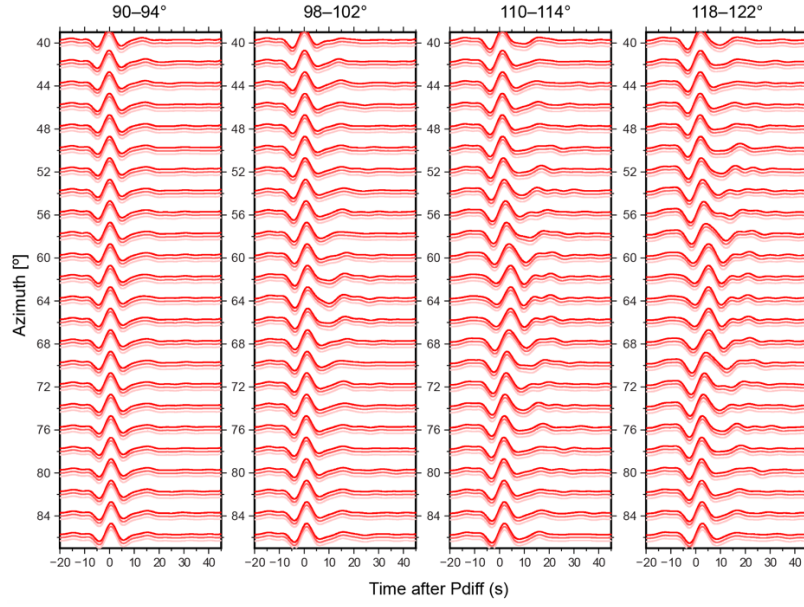


fig. S9. Synthetic Pdiff waveforms. Comparison of synthetic waveforms for a cylindrical ULVZ structure (as fig. S6), but with varying density (pale red: $\Delta\rho = +0\%$, medium red: $\Delta\rho = +10\%$, bright red: $\Delta\rho = +15\%$). Waveforms are computed with SPECfEM-3D. Waveforms are computed using SPECfEM-3D and are slightly offset by azimuth for clarity. Differences between the waveforms are negligible.

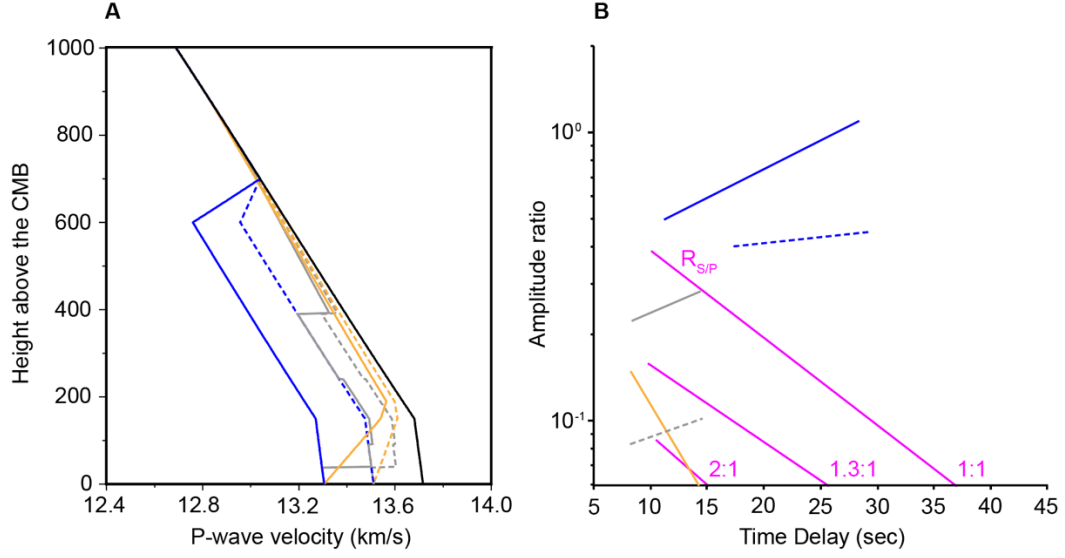


fig. S10. Predicted Pdiff postcursor delay times and log-amplitude ratios for three regional 1-D models. (A) P-wave velocity profiles used for waveform simulations. (B) Corresponding synthetic delay times and log-amplitude measurements (gray: 33; blue: 34; orange: 35). The velocity profiles are derived from original S-wave velocity models, modified using $R_{S/P} = 1:1$ (solid lines) and 2:1 (dashed lines). Magenta lines indicate predictions from the cylindrical ULVZ model with $H = 50$ km with varying $R_{S/P}$ (Fig. 3B). Synthetic Pdiff postcursors from the red dashed model are too weak to be plotted.

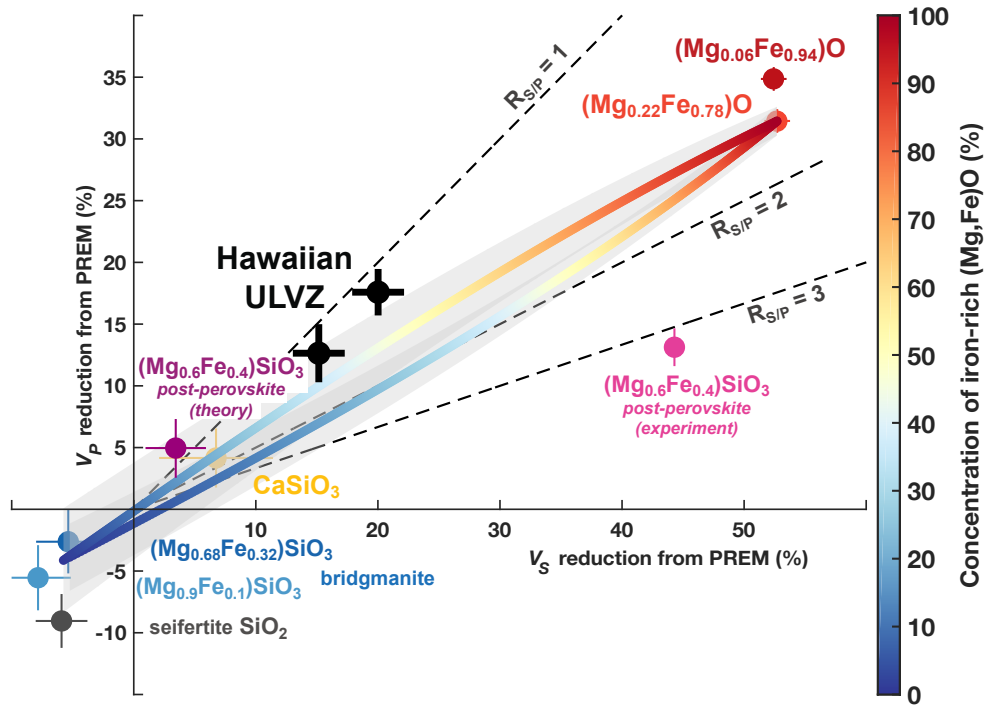


fig. S11. Effect of lower iron content on bulk rock sound velocities. Shaded regions indicate uncertainties on the aggregate sound velocities. Sound velocity values and references for individual phases are listed in table S1.

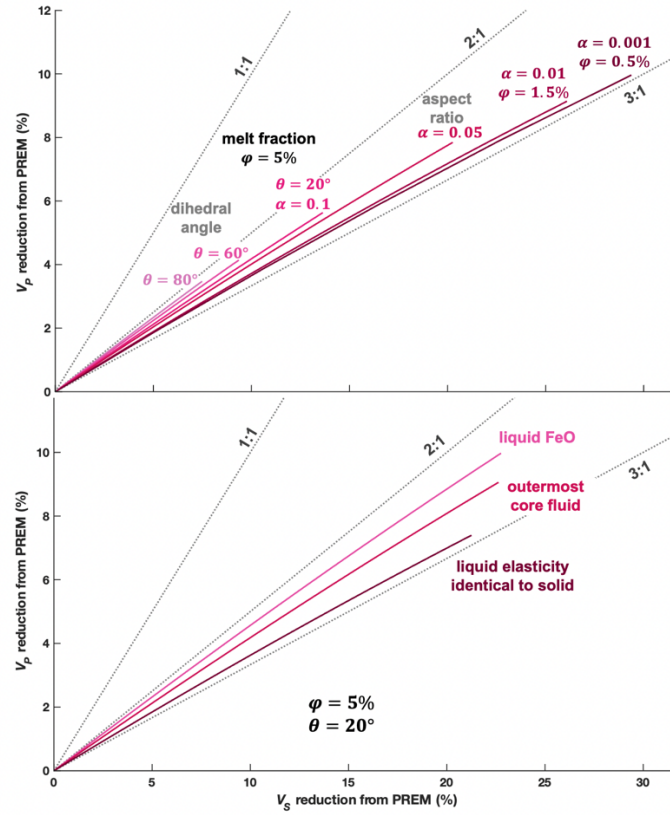


fig. S12. Sound velocity reductions in partially molten rock. Forward models showcase examples of how melt geometry, fraction, and composition affect the relative reductions in P and S wave speeds.

table. S1. Mineral and rock properties used for mineralogical models. Values are previously calculated for core-mantle boundary conditions (38,59) or taken from published studies (64–67).

Phase	V_P (km/s)	V_S (km/s)	ρ (g/cm ³)
(Mg _{0.22} Fe _{0.78})O magnesiowüstite “Mw78”	9.40 ± 0.14	3.44 ± 0.09	7.89 ± 0.02
(Mg _{0.06} Fe _{0.94})O magnesiowüstite “Mw94”	8.93 ± 0.14	3.46 ± 0.09	8.54 ± 0.02
(Mg _{0.90} Fe _{0.10})SiO ₃ bridgmanite “Br10”	14.47 ± 0.36	7.84 ± 0.20	5.42 ± 0.05
(Mg _{0.68} Fe _{0.32})SiO ₃ bridgmanite “Br32”	14.07 ± 0.35	7.66 ± 0.19	5.73 ± 0.06
CaSiO ₃	13.14 ± 0.33	6.78 ± 0.34	5.49 ± 0.05
(Mg _{0.60} Fe _{0.40})SiO ₃ post-perovskite (experiment)	11.91 ± 0.21	4.05 ± 0.03	5.90 ± 0.06
(Mg _{0.60} Fe _{0.40})SiO ₃ post-perovskite (theory)	13.03 ± 0.33	7.02 ± 0.18	5.90 ± 0.06
SiO ₂ seifertite	14.95 ± 0.30	7.70 ± 0.15	5.45 ± 0.10
Mid-ocean ridge basalt (MORB)	13.85 ± 0.15	7.42 ± 0.07	5.62 ± 0.11

table. S2. Properties of iron-rich mineral assemblages that reproduce the seismic models.

The four compositions are identical to those plotted in the inset of Fig. 5 and represent approximate bounds on compositions that exhibit sound velocities compatible with seismic models, within uncertainties. $R_{S/P}$ values for each composition are listed as ranges that account for uncertainties in the seismic velocities of the mineral assemblage, as showcased by the shaded lines in Fig. S12.

Property	Mw94+Br32 (Reuss, h = 75 km)	Mw94+Br32 (Voigt, h = 50 km)	Mw78+Br10 (Voigt, h = 75 km)	Mw78+Br10 (Voigt, h = 50 km)
X_{Br} (%)	84	50	67	46
X_{Mw} (%)	16	50	33	54
$\delta \ln \rho$ (%)	10	28	12	21
$R_{S/P}$	1 – 2.4	1.2 – 1.7	1.2 – 2.3	1.3 – 1.9
FeO (mol%)	28.5	55	29.1	44.4
MgO (mol%)	29.5	20	37.4	32.6
SiO ₂ (mol%)	42	25	33.5	23

Data S1. (separate file)

Table of earthquake events used in the study.

Data S2. (separate file)

List of seismic networks providing data, with corresponding citations.

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